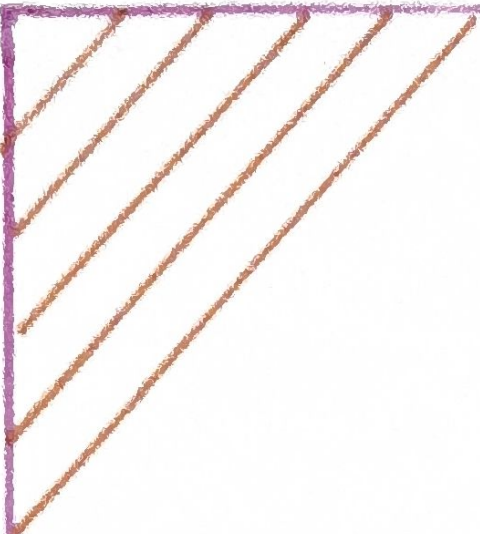




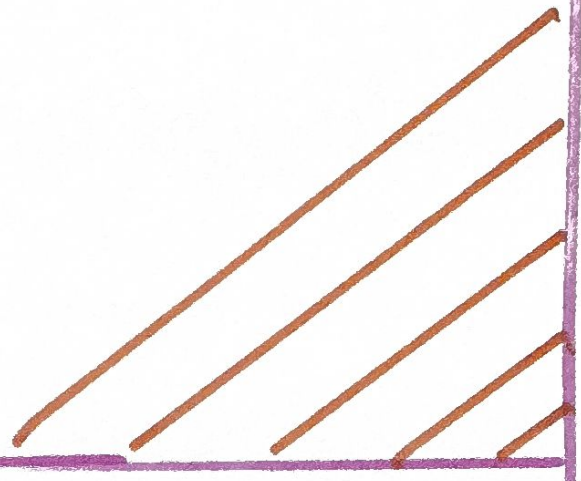
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Course Code: ITA0501

Course: Computer Vision

Assessment: Col-ATI-Short Answer



1. Sampling - low spatial Resolution

The problem is Insufficient sampling, so small objects are not captured with enough pixels. Increase the sampling rate / spatial resolution by using a higher-resolution camera. This provides more pixels and makes small objects clearer.

2) Image Enhancement - Dark and Noisy Night Image

Apply histogram equalization to improve brightness and contrast. A suitable noise-reduction filter can also be used. This makes important details more visible while reducing unwanted noise.

3) Quantization - Medical Image

Quantization converts continuous intensity values into a fixed number of digital intensity levels. For example, an image can use 256 levels (0-255).

This allows the continuous medical image to be stored and processed digitally.

4) Image Enhancement - uneven lighting

Apply Contrast enhancement or adaptive histogram equalization (CLAHE). It improves local contrast in different areas of the image, making defects easier to see even when lighting is uneven.

5) Image Segmentation - Self Driving Car

Use image segmentation to divide the image into regions such as road, vehicles, pedestrians and background. This helps the car understand its surroundings and make decisions such as steering, braking, and avoiding obstacles.

6) Edge detection - Crack Detection

Apply an edge detection technique such as the

Canny edge detector. Cracks usually locate sharp intensity changes, which appear as edges. Detecting the edges helps identify the location and shape of cracks.

7) Filtering - Noisy Medical Image

Apply a Median filter to remove noise while preserving important edges. This improves image quality and makes the boundaries of a tumor easier to identify.

8) Object Detection - Traffic Monitoring

Use an object detection model to identify vehicles and locate them using bounding boxes. Vehicle tracking can then follow their movement. This helps monitor traffic flow, detect congestion, and improve traffic management.

9) Motion Blur - Moving Object

The blur is mainly caused by motion during image Acquisition producing motion blur. use deblurring / restoration technique to reduce the blur and recover sharper image details.

10) Image Classification - Quality Inspection

Train an Image classification model using image of good and defective products. The model classifies each new product as "Good" or "Defective". This support fast, consistent and automated quality inspection